Comparar com NULL

```
SELECT *
FROM population
WHERE tot_pop = NULL;
```

e

```
SELECT *
FROM population
WHERE tot_pop = NULL;
```

Não trazem nenhum resultado!

Operador IS NULL

```
SELECT *
FROM population
WHERE tot_pop IS NULL;
```

	ABC country	123 ref_year	123 tot_pop
1	Holy See	2,001	[NULL]
2	Holy See	2,002	[NULL]
3	Holy See	2,003	[NULL]
4	Holy See	2,004	[NULL]
5	Holy See	2,005	[NULL]
6	Holy See	2,006	[NULL]
7	Holy See	2,007	[NULL]
8	Holy See	2,008	[NULL]
9	Holy See	2,009	[NULL]
10	Holy See	2,010	[NULL]

Operador IS NOT NULL

```
SELECT *
FROM population
WHERE tot_pop IS NOT NULL;
```

	country	ref_year	tot_pop
1	Afghanistan	1,800	3,280,000
2	Afghanistan	1,801	3,280,000
3	Afghanistan	1,802	3,280,000
4	Afghanistan	1,803	3,280,000
5	Afghanistan	1,804	3,280,000
6	Afghanistan	1,805	3,280,000
7	Afghanistan	1,806	3,280,000
8	Afghanistan	1,807	3,280,000
9	Afghanistan	1,808	3,280,000
10	Afghanistan	1,809	3,280,000

Operador BETWEEN

```
SELECT country, ref_year, tot_deaths
FROM child_mortality
WHERE tot_deaths BETWEEN 570 AND 630;
```

	ABC country	123 ref_year	123 tot_deaths
30	Barbados	1,919	572
31	Barbados	1,923	598
32	Cuba	1,896	575
33	Cuba	1,897	589
34	Cuba	1,898	604
35	Cuba	1,899	621
36	Ireland	1,848	587
37	Ireland	1,849	598
38	Iceland	1,846	608
39	Iceland	1,882	588
40	Pakistan	1,921	570
41	Ukraine	1,943	577

Operador BETWEEN e datas

```
SELECT "date", open, high, low, close, "adj close", volume
FROM petrobras
WHERE "date" BETWEEN '2020-02-20' AND '2020-03-20';
```

	Date	Open	High	Low	Close	Adj Close	Volume
1	2020-02-20	14.77	14.93	14.37	14.4	5.303012	21,453,600
2	2020-02-21	14.17	14.18	13.89	14.03	5.166753	22,637,200
3	2020-02-24	13.14	13.28	12.71	13.08	4.816903	38,318,000
4	2020-02-25	13.2	13.3	12.62	12.82	4.721154	26,389,700
5	2020-02-26	12.84	13.1	12.45	12.55	4.621723	41,263,500
6	2020-02-27	12	12.55	11.71	12.08	4.448638	41,795,900
7	2020-02-28	11.56	12.11	11.51	12.1	4.456003	46,881,600
8	2020-03-02	12.2	12.57	12.02	12.49	4.599626	29,931,200
9	2020-03-03	12.51	12.91	12	12.16	4.478099	37,360,900
10	2020-03-04	12.45	12.45	12.1	12.33	4.540704	23,901,800
11	2020-03-05	12.01	12.05	11.31	11.74	4.323427	42,527,900
12	2020-03-06	10.7	10.87	10.27	10.52	3.874145	52,078,200
13	2020-03-09	7.55	8.26	7.04	7.26	2.673602	95,745,000
14	2020-03-10	8.5	8.57	7.35	8.01	2.9498	54,755,700
15	2020-03-11	7.52	7.77	6.64	6.85	2.522614	55,715,500
16	2020-03-12	5.75	5.89	4.83	5.45	2.007043	86,571,700
17	2020-03-13	6.69	6.74	5.57	6.58	2.423182	59,809,100
18	2020-03-16	5.4	5.96	5.13	5.25	1.93339	35,275,300
19	2020-03-17	5.41	5.66	5.14	5.4	1.988629	51,896,200
20	2020-03-18	4.56	4.82	4.01	4.31	1.587221	49,017,000
21	2020-03-19	4.4	5.12	4.1	5.06	1.863419	56,339,600
22	2020-03-20	5.34	5.39	4.72	4.85	1.786084	38,227,800

Operadores lógicos

operador lógico	significado
AND	e
OR	ou
NOT	negação

AND – tabela verdade

AND	verdadeiro	falso
verdadeiro	verdadeiro	falso
falso	falso	falso

Exemplo de uso de AND

```
SELECT country, co2_pc
FROM co2_emissions_pc
WHERE ref_year = 2020
  AND co2_pc > 20.0;
```

	ABC country	123 co2_pc
1	Brunei	22.4
2	Kuwait	22.1
3	Qatar	25.6
4	Singapore	24.1

OR – tabela verdade

OR	verdadeiro	falso
verdadeiro	verdadeiro	verdadeiro
falso	verdadeiro	falso

Exemplo de uso de OR

```
SELECT country, wb_regions
FROM country
WHERE wb_regions = 'North America'
  OR wb_regions = 'Latin America & Caribbean';
```

	country	wb_regions
25	Paraguay	Latin America & Caribbean
26	Peru	Latin America & Caribbean
27	St. Kitts and Nevis	Latin America & Caribbean
28	St. Lucia	Latin America & Caribbean
29	St. Vincent and the Gr	Latin America & Caribbean
30	Suriname	Latin America & Caribbean
31	Trinidad and Tobago	Latin America & Caribbean
32	Uruguay	Latin America & Caribbean
33	Venezuela	Latin America & Caribbean
34	Canada	North America
35	USA	North America

NOT – tabela verdade

NOT	verdadeiro	falso
	falso	verdadeiro

Exemplo de uso de NOT

```
SELECT country, wb_regions
FROM country
WHERE NOT (wb_regions = 'North America'
  OR wb_regions = 'Latin America & Caribbean');
```

	country	wb_regions
1	Afghanistan	South Asia
2	Albania	Europe & Central Asia
3	Algeria	Middle East & North Africa
4	Andorra	Europe & Central Asia
5	Angola	Sub-Saharan Africa
6	Armenia	Europe & Central Asia
7	Australia	East Asia & Pacific
8	Austria	Europe & Central Asia
9	Azerbaijan	Europe & Central Asia
10	Bahrain	Middle East & North Africa

Encadeando expressões lógicas – resultado inesperado

```
SELECT country, ref_year, tot_deaths
FROM child_mortality
WHERE tot_deaths BETWEEN 57 AND 63
  AND country = 'China' OR country = 'Brazil';
```

A expressão é avaliada da esquerda para direita.

	country	ref_year	tot_deaths
292	Brazil	2,091	3.23
293	Brazil	2,092	3.19
294	Brazil	2,093	3.15
295	Brazil	2,094	3.11
296	Brazil	2,095	3.07
297	Brazil	2,096	3.03
298	Brazil	2,097	3
299	Brazil	2,098	2.96
300	Brazil	2,099	2.92
301	Brazil	2,100	2.92
302	China	1,980	62.5
303	China	1,981	59.7
304	China	1,982	57.5

Valores fora da faixa desejada!

Encadeando expressões lógicas – expressão correta

```
SELECT country, ref_year, tot_deaths
FROM child_mortality
WHERE tot_deaths BETWEEN 57 AND 63
  AND (country = 'China' OR country = 'Brazil');
```

	ABC country	123 ref_year	123 tot_deaths
1	Brazil	1,990	63
2	Brazil	1,991	60.2
3	Brazil	1,992	57.2
4	China	1,980	62.5
5	China	1,981	59.7
6	China	1,982	57.5

O uso de parênteses muda a ordem de precedência.

Operador IN

```
SELECT country, ref_year, tot_deaths
FROM child_mortality
WHERE tot_deaths BETWEEN 57 AND 63
  AND country IN ('China', 'Brazil');
```

	ABC country	123 ref_year	123 tot_deaths
1	Brazil	1,990	63
2	Brazil	1,991	60.2
3	Brazil	1,992	57.2
4	China	1,980	62.5
5	China	1,981	59.7
6	China	1,982	57.5

O resultado é o mesmo da *query* anterior.

Operador IN substitui vários OR

```sql
SELECT country, co2_pc
FROM co2_emissions_pc
WHERE ref_year = 2022
  AND (country = 'Canada'
   OR  country = 'France'
   OR  country = 'Germany'
   OR  country = 'Italy'
   OR  country = 'Japan'
   OR  country = 'UK'
   OR  country = 'USA');
```

```sql
SELECT country, co2_pc
FROM co2_emissions_pc
WHERE ref_year = 2022
  AND country IN ('Canada', 'France', 'Germany',
                  'Italy', 'Japan', 'UK', 'USA');
```

	ABC country	123 co2_pc
1	Canada	13.2
2	Germany	9.3
3	France	6.07
4	UK	6.93
5	Italy	6.92
6	Japan	9.99
7	USA	16.4

Operador LIKE no início da frase

```
SELECT country, mean_babies
FROM fertility
WHERE ref_year = 2021
  AND country LIKE 'Congo%';
```

	ABC country	123 mean_babies
1	Congo, Dem. Rep.	5.62
2	Congo, Rep.	4.37

Operador LIKE no final da frase

```
SELECT country, mean_babies
FROM fertility
WHERE ref_year = 2021
  AND country LIKE '%land';
```

	ABC country	123 mean_babies
1	Switzerland	1.56
2	Finland	1.79
3	Greenland	[NULL]
4	Ireland	1.97
5	Iceland	1.89
6	New Zealand	1.94
7	Poland	1.29
8	Thailand	1.42

Operador LIKE em qualquer lugar da frase

```
SELECT country, mean_babies
FROM fertility
WHERE ref_year = 2021
  AND country LIKE '%Guinea%';
```

	ABC country	123 mean_babies
1	Guinea	4.48
2	Guinea-Bissau	4.27
3	Equatorial Guinea	4.26
4	Papua New Guinea	3.43



FGV EESP

Todos os direitos reservados. Textos, vídeos, sons, imagens, gráficos e demais componentes deste material são protegidos por direitos autorais e outros direitos de propriedade intelectual, de forma que é proibida a reprodução no todo ou em parte, sem a devida autorização.

Como citar este material:
BONATO, Antonio Sergio Ferreira. **Operadores especiais e lógicos**. Rio de Janeiro: FGV, 2024.